



Exercise 1

Create an artificial data set with three columns named A, B, and C.

1. Column A contains 10 random numbers sampled from a Bernoulli distribution with rate 0.3; column B contains 10 random numbers from the uniform distribution over $[0, 1]$; and column C contains 10 samples from the set ("hip", "hop"), with equal probability for "hip" and "hop".
2. Create a vector whose i -th element is the sum of the i -th entries of columns A and B of the data frame created in part 1.
3. Select all rows with "hop" in column C and display the resulting data frame.

Exercise 2

OpenML, CSV files, and visualization

In this exercise, you will load some data, write it to a CSV file, and visualize it.

1. Look at the description of OpenML dataset 61. Hint: use the OpenML documentation.
2. Load dataset 61 from OpenML into a pandas data frame.
3. Write the data to a CSV file on your hard disk. Hint: use the `to_csv` function.
4. Load the data in the CSV file as a data frame. Hint: use `pandas pd.read_csv`. (Saving the Iris data set and loading it immediately again is intentionally redundant here; the goal is to practice file I/O.)
5. Produce a scatter plot showing sepal-length versus sepal-width. Make sure both axes have the correct labels.
6. Add the identity line $y = x$ to the same plot in red.

Exercise 3

For each scenario, explain whether the problem is classification or regression, indicate whether we are mainly interested in interpretation or accurate prediction, and give n (the number of samples) and p (the number of dimensions of the input).

- We collect data on the top 500 firms in the US. For each firm, we record profit, number of employees, industry, and CEO salary. We are interested in understanding which factors affect CEO salary.
- We are considering launching a new product and wish to know whether it will be a success or a failure. We collect data on 20 similar products that were previously launched. For each product, we record whether it was a success or failure, the price charged, the marketing budget, the competition price, and ten other variables.
- We are interested in predicting the percentage change in the USD/Euro exchange rate in relation to weekly changes in world stock markets. We collect weekly data for all of 2012. For each week, we record the percentage change in USD/Euro, the percentage change in the US market, the percentage change in the British market, and the percentage change in the German market.